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Gargantua's Anatomy

If we know the mass of a black hole and how fast it spins, then from Einstein's relativistic laws we can deduce all the hole's other properties: its size, the strength of its gravitational pull, how much its event horizon is stretched outward near the equator by centrifugal forces, the details of the gravitational lensing of objects behind it. Everything.

This is amazing. So different from everyday experience. It is as though knowing my weight and how fast I can run, you could deduce everything about me: the color of my eyes, the length of my nose, my IQ, . . .

John Wheeler (my mentor, who gave "black holes" their name) has described this by the phrase "A black hole has no hair"—no extra, *independent* properties beyond its mass and its spin. Actually, he should have said, "A black hole has only two hairs, from which you can deduce everything else about it," but that's not as catchy as "no hair," which quickly became embedded in black-hole lore and scientists' lexicon.[13]